Dynamic Partial Cointegration & Neural Kalman Filtering for Statistical Arbitrage

A Unified Study: State-Space Modelling, Partial Cointegration, Pair Screening, KalmanNet-Learned Gains, Differentiable Sharpe Training, and Out-of-Sample Evaluation

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Abstract

Pairs trading profits from the short-term divergence of two related assets, but fixed hedge ratios and durable drift in the spread undermine classical implementations; this report is constructing a Dynamic Partial Cointegration (Dynamic-PCI) framework that's modeling the hedge ratio as a hidden random walk and splits the spread into a mean-reverting and a random-walk component, and replaces the analytical Kalman gain by a GRU network (KalmanNet) trained with a two-stage curriculum that's ending in a cost-aware differentiable Sharpe loss; this model is generally benchmarked against static baselines on 14 pairs over four out-of-sample years (2017–2021) with 10 bps transaction costs, in actual application.

1 Mathematical and Statistical Foundations

Pairs trading [10] rests on cointegration [9, 13, 11, 12]. Learning the filter gain from data, as in KalmanNet [4, 2], tackles the Gaussian-noise assumption (Section 4); this section collects the statistical tools employed throughout; two facts are weakening the textbook version, such as the equilibrium hedge ratio drifts over time. And spreads include a durable as well as a temporary component, formalised as partial cointegration (PCI) by Clegg and Krauss [1]; Yu [6] monitors the hedge ratio with a Kalman filter, at least in practice.

1.1 Integration Orders and Cointegration

Definition 1.1 (Integration order)**.** What's more, a scalar series x_t is blended of order d, written $I(d)$, if $\Delta^d x_t$ is weakly stationary while $\Delta^{d-1} x_t$ is not, in real-world scenarios.

A random walk $R_t = R_{t-1} + \varepsilon_t$ is $I(1)$ with $\mathrm{Var}(R_t) = t\sigma^2$ expanding without bound; an AR(1) process $M_t = \rho M_{t-1} + \varepsilon_t$ with $|\rho| < 1$ is $I(0)$ with stationary variance $\sigma^2/(1-\rho^2)$; so, as well, log or level prices are typically $I(1)$, to a significant degree.

Definition 1.2 (Cointegration [9])**.** Two $I(1)$ series $X_{1,t}, X_{2,t}$ are, indeed, cointegrated if there's β with $S_t = X_{2,t} - \beta X_{1,t}$ stationary.

The Engle–Granger two-step procedure regresses X_2 on X_1 by OLS and tests the residual for a unit root, broadly speaking. A well-known subtlety is that OLS chooses β to make the residual as small as possible, so the residual looks "more stationary" than a series with a pre-specified β; so the usual Dickey–Fuller critical values are too lenient and residual-based tests are needing adjusted (more negative) critical values [9], all things considered. We return to this when construing screening p-values in Section 3.

1.2 Partial Cointegration

Definition 1.3 (Partial cointegration [1])**.** The pair is partially cointegrated if

$$X_{2,t} - \beta X_{1,t} = M_t + R_t, \qquad M_t = \rho M_{t-1} + \varepsilon_t^M, \quad R_t = R_{t-1} + \varepsilon_t^R, \tag{1}$$

with $|\rho| < 1$ and autonomous Gaussian innovations $\varepsilon_t^M \sim \mathcal{N}(0, \sigma_M^2)$, $\varepsilon_t^R \sim \mathcal{N}(0, \sigma_R^2)$.

The concrete question is usually how much of the spread's change is temporary, so a pure random-walk spread is $\sigma_M = 0$ (or $\rho = 1$), at least for now; standard cointegration is the special case $\sigma_R = 0$.

Proposition 1.4 (Variance ratio R^2_{MR} and half-life)**.** *Under* (1)*, the first differences* ΔM_t *and* ΔR_t *are normally stationary with*

$$\operatorname{Var}(\Delta M_t) = \frac{2\sigma_M^2}{1+\rho}, \qquad \operatorname{Var}(\Delta R_t) = \sigma_R^2,$$

Clearly, right so that the share of mean-reverting variance in the differenced spread is

$$R^2_{\mathrm{MR}} = \frac{\operatorname{Var}(\Delta M_t)}{\operatorname{Var}(\Delta M_t) + \operatorname{Var}(\Delta R_t)} = \frac{2\sigma_M^2}{2\sigma_M^2 + (1+\rho)\sigma_R^2}. \tag{2}$$

What's more, $\mathbb{E}[M_{t+h} \mid M_t] = \rho^h M_t$*, so the half-life, i.e. the horizon at which the anticipated deviation halves, is*

$$\mathrm{HL} = -\frac{\ln 2}{\ln|\rho|}. \tag{3}$$

Proof. Seeing as $\operatorname{Var}(M_t) = \gamma_0 = \sigma_M^2/(1-\rho^2)$ and $\operatorname{Cov}(M_t, M_{t-1}) = \rho\gamma_0$, we get $\operatorname{Var}(\Delta M_t) = 2\gamma_0(1-\rho) = 2\sigma_M^2/(1+\rho)$. What's more $\Delta R_t = \varepsilon_t^R$. Looping the AR(1) recursion gives $\mathbb{E}[M_{t+h} \mid M_t] = \rho^h M_t$; setting $\rho^h = \frac{1}{2}$ generates (3) for the half-life. Swapping is giving (2). □

Proposition 1.5 (Reduced form and weak identification)**.** *Let* $Y_t = M_t + R_t$ *with* M_t, R_t *as in* (1)*, and then an ARMA(1,1) process is being followed by* ΔY_t*,*

$$(1-\rho L)\,\Delta Y_t = (1-L)\varepsilon_t^M + (1-\rho L)\varepsilon_t^R,$$

And whose right-hand side is an MA(1). As well, as $\rho \to 1$*, the autoregressive and moving-average roots approach cancellation,* M_t *becomes a random walk, at least empirically; in fact, as, well,* σ_M^2 *and* σ_R^2 *enter the likelihood only through the their sum.*

Argument. Use $(1-\rho L)(1-L)$ to Y_t and use $(1-\rho L)M_t = \varepsilon_t^M$, $(1-L)R_t = \varepsilon_t^R$. At $\rho = 1$ the noise enters only, through $\sigma_M^2 + \sigma_R^2$, so near $\rho = 1$ the likelihood is nearly flat along the direction trading σ_M^2 against σ_R^2, at least in principle; the finding is an MA(1) right-hand side with three reduced-form quantities, so the model is exactly spotted, in the main. □

Remark 1.6. This is an val, not a full asymptotic finding, and is uniform with $\hat{\rho}$ hitting 0.999 on long windows (Section 9.1) and with σ_R being recovered less accurately than ρ in modelling (Section 7), in many respects.

1.3 Linear Gaussian State-Space Models and the Kalman Filter

A linear Gaussian state-space model has hidden state $\boldsymbol{z}_t \in \mathbb{R}^m$ and observation $\boldsymbol{y}_t \in \mathbb{R}^n$:

$$\boldsymbol{z}_t = \boldsymbol{F}\boldsymbol{z}_{t-1} + \boldsymbol{w}_t,\ \boldsymbol{w}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}); \qquad \boldsymbol{y}_t = \boldsymbol{H}_t\boldsymbol{z}_t + \boldsymbol{v}_t,\ \boldsymbol{v}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}_{\mathrm{obs}}). \tag{4}$$

The Kalman filter [8, 15] computes $\hat{\boldsymbol{z}}_{t|t} = \mathbb{E}[\boldsymbol{z}_t \mid \boldsymbol{y}_{1:t}]$ recursively:

$$\hat{\boldsymbol{z}}_{t|t-1} = \boldsymbol{F}\hat{\boldsymbol{z}}_{t-1|t-1}, \qquad \boldsymbol{P}_{t|t-1} = \boldsymbol{F}\boldsymbol{P}_{t-1|t-1}\boldsymbol{F}^{\top} + \boldsymbol{Q}, \tag{5}$$

$$\boldsymbol{e}_t = \boldsymbol{y}_t - \boldsymbol{H}_t\hat{\boldsymbol{z}}_{t|t-1}, \qquad \boldsymbol{S}_t = \boldsymbol{H}_t\boldsymbol{P}_{t|t-1}\boldsymbol{H}_t^{\top} + \boldsymbol{R}_{\mathrm{obs}}, \tag{6}$$

$$\boldsymbol{K}_t = \boldsymbol{P}_{t|t-1}\boldsymbol{H}_t^{\top}\boldsymbol{S}_t^{-1}, \tag{7}$$

$$\hat{\boldsymbol{z}}_{t|t} = \hat{\boldsymbol{z}}_{t|t-1} + \boldsymbol{K}_t\boldsymbol{e}_t, \qquad \boldsymbol{P}_{t|t} = (\boldsymbol{I} - \boldsymbol{K}_t\boldsymbol{H}_t)\boldsymbol{P}_{t|t-1}. \tag{8}$$

Theorem 1.7 (Optimality and innovations [8, 14])**.** $\hat{\boldsymbol{z}}_{t|t}$ *As, well, the* standardised innovations $\boldsymbol{S}_t^{-1/2}\boldsymbol{e}_t$ *are i.i.d.* $\mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_n)$*. is the minimum mean-squared-error estimator of* $\boldsymbol{z}_t$ *given* $\boldsymbol{y}_{1:t}$*, if* (4) *holds with known coefficients, on the whole.*

The exact likelihood and a diagnostic are given by the innovation property - with a correct model, standardised innovations are white and normal (tested in Section 7), in the main; the gain (7) is the only place where the noise covariances enter the update, so that said, this is what the neural network replaces (Section 4), at least for now.

1.4 Maximum Likelihood by Prediction-Error Decomposition

By the chain rule, $p(\boldsymbol{y}_{1:T} \mid \boldsymbol{\theta}) = \prod_t p(\boldsymbol{y}_t \mid \boldsymbol{y}_{1:t-1}; \boldsymbol{\theta})$. What's more, $\boldsymbol{y}_t = \boldsymbol{H}_t \boldsymbol{z}_t + \boldsymbol{v}_t$ is linear in it plus autonomous noise, so $\boldsymbol{y}_t \mid \boldsymbol{y}_{1:t-1} \sim \mathcal{N}(\boldsymbol{H}_t \hat{\boldsymbol{z}}_{t|t-1}, \boldsymbol{S}_t)$ with density $(2\pi)^{-n/2}|\boldsymbol{S}_t|^{-1/2} \exp(-\frac{1}{2}\boldsymbol{e}_t^\top \boldsymbol{S}_t^{-1}\boldsymbol{e}_t)$, in real-world scenarios. Given $\boldsymbol{y}_{1:t-1}$ the state is currently $\mathcal{N}(\hat{\boldsymbol{z}}_{t|t-1}, \boldsymbol{P}_{t|t-1})$. Taking logs and summing gives

$$\ell(\boldsymbol{\theta}) = -\tfrac{1}{2} \sum_{t=1}^{T} \Big(\ln |2\pi \boldsymbol{S}_t| + \boldsymbol{e}_t^\top \boldsymbol{S}_t^{-1} \boldsymbol{e}_t \Big), \tag{9}$$

which for the production model ($n = 1$) is $-\frac{1}{2}\sum_t [\ln(2\pi S_t) + e_t^2/S_t]$: the first term penalises predictive uncertainty, so one filter pass per $\boldsymbol{\theta}$ checks ℓ in $O(T)$, in actual application. L-BFGS-B [18] are employed on the formation window by maximisation. And the second penalises surprises relative to it, and the factors are $\boldsymbol{\theta}_{\text{dyn}} = [\beta_0, \rho, \ln \sigma_M, \ln \sigma_R, \ln q_\beta]$ (Dynamic-PCI) and $[\beta_0, \rho, \ln \sigma_M, \ln \sigma_R]$ with $q_\beta^2 \equiv 10^{-12}$ (Static-PCI); logs are keeping variances positive, in actual application.

1.5 Unit-Root Testing: the ADF Test

Idea. In $s_t = \phi s_{t-1} + \varepsilon_t$, $\phi = 1$ is a random walk and $|\phi| < 1$ is mean-reverting. Writing $\Delta s_t = \gamma s_{t-1} + \varepsilon_t$ with $\gamma = \phi - 1$, the question is whether $\gamma = 0$ or $\gamma < 0$.

Test. The Augmented Dickey–Fuller test [16, 17] fits $\Delta s_t = a + \gamma s_{t-1} + \sum_{i=1}^{p} \delta_i \Delta s_{t-i} + u_t$ by OLS (lagged differences absorb serial correlation) and tests $H_0 : \gamma = 0$ against $\gamma < 0$ using $\hat{\gamma}/\text{se}(\hat{\gamma})$, which follows the non-standard Dickey–Fuller distribution rather than a t distribution. A small p-value is evidence against a unit root; a large one means no evidence of stationarity, not proof of non-stationarity.

Use and caveats. The screen applies it to the static spread $X_{2,t} - \beta_0 X_{1,t}$ and to the filtered state $\hat{M}_t$. The test has low power near a unit root ($\rho = 0.995$ has a half-life of 138 days); with an estimated β the standard critical values are anti-conservative [9]; and $\hat{M}_t$ is built assuming $\rho < 1$, so a small p on it is not independent confirmation of cointegration.

1.6 Sampling Distribution of the Sharpe Ratio

The annualised Sharpe ratio is often $\widehat{SR} = \sqrt{252}\,\bar{r}/s_r$ [22]. Under i.i.d. returns, Lo [23] indicates for n_y years of data

$$\text{SE}(\widehat{SR}) \approx \sqrt{\frac{1 + \frac{1}{2}\widehat{SR}^{\,2}}{n_y}}. \tag{10}$$

With the 1,007-day out-of-sample window, $n_y \approx 4.0$, so even a Sharpe of zero has a standard deviation of ≈ 0.50: a 95% interval is roughly ± 1.0 Sharpe units, at least for now. Don't forget, scrutinized probably, the best gathered Sharpe is also upward-biased by selection; Bailey and López de Prado [24] formalise this as the *deflated Sharpe ratio*, when many model–pair combinations are; this is the central reason the pair-level Sharpe ratios in Section 8 should be read as descriptive, at least in practice.

2 The Dynamic-PCI State-Space Model

2.1 Economic Assumptions and Asset Ordering

Let $X_{1,t}$ mean the *driver* (the larger, more liquid asset) and $X_{2,t}$ the *follower*; in turn, exchanging the functions gives $\beta' \approx 1/\beta$ and a diverse noise structure, in the main, and a unstable, illiquid driver would also destabilise β_t, so the larger asset is always X_1, in actual application. Trades are supposed not to move prices, as is often standard in pairs-trading studies [2, 1], and to keep β_t from absorbing high-frequency deviations that belong to M_t, the hedge-ratio noise q_β is mainly kept small relative to σ_M (*scale separation*): β_t should represent only low-frequency, macro-level drift. Prices are currently normalised to a base of 1.0 on the first formation day so that all filter quantities are $O(1)$ and scale-free (this matters for numerical robustness, Section 5.4).

2.2 The Production 3-State Model

The production model treats the driver price as a known *exogenous regressor* and monitors only the three genuinely unobservable magnitudes,

$$\boldsymbol{z}_t = \begin{bmatrix} \beta_t \\ M_t \\ R_t \end{bmatrix}, \quad \boldsymbol{F} = \mathrm{diag}(1, \rho, 1), \quad \boldsymbol{Q} = \mathrm{diag}(q_\beta^2, \sigma_M^2, \sigma_R^2), \tag{11}$$

with the scalar finding

$$X_{2,t} = \boldsymbol{H}_t \boldsymbol{z}_t + v_t, \qquad \boldsymbol{H}_t = [\, X_{1,t},\ 1,\ 1\,], \qquad v_t \sim \mathcal{N}(0, \sigma_{\mathrm{obs}}^2). \tag{12}$$

Bluntly, this is the partial-cointegration equation $X_{2,t} = \beta_t X_{1,t} + M_t + R_t + v_t$ written in state-space form; indeed, as $\boldsymbol{H}_t$ includes $X_{1,t}$ maybe, the filter is linear in the state but not time-invariant, in real-world scenarios. The innovation variance $S_t = \boldsymbol{H}_t \boldsymbol{P}_{t|t-1} \boldsymbol{H}_t^\top + \sigma_{\mathrm{obs}}^2$ is a positive *scalar*, so the gain is needing a division rather than a matrix inversion, at least in principle.

2.3 A Glimpse of the 4-State Model and Why It Was Discarded

The driver price was, more accurately, placed inside the state vector by an earlier representation,

$$\boldsymbol{z}_t = \begin{bmatrix} X_{1,t} \\ \beta_t \\ M_t \\ R_t \end{bmatrix}, \quad \boldsymbol{F} = \mathrm{diag}(1, 1, \rho, 1), \quad \boldsymbol{Q} = \mathrm{diag}(\sigma_X^2, q_\beta^2, \sigma_M^2, \sigma_R^2), \quad \boldsymbol{H}_t = \begin{bmatrix} 0 & X_{1,t} & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}, \tag{13}$$

The second row simply re-observes the driver. With observation $\boldsymbol{y}_t = [X_{2,t}, X_{1,t}]^\top$ and a 2×2 noise matrix $\boldsymbol{R}_{\mathrm{obs}}$, and the 4-state model was dropped for five reasons, at least empirically.

1. **An observable does not belong in the latent state** [15, 14]: $X_{1,t}$ is gathered exactly, so estimating it's unnecessary.
2. **Near-singular innovation covariance.** $\boldsymbol{S}_t$ is normally singular or ill-conditioned and needs an artificial floor ϵ to invert with zero noise on the second row, at least in practice.
3. **Spurious cross-covariance** between X_1 and (β, M, R), so driver errors leak into $\hat{\beta}_t$ and $\hat{M}_t$.
4. **Signal collapse.** What's more, the extra factor let the optimiser inflate q_β^2 so that β_t absorbed the temporary spread; for KO–PEP it estimated $R^2_{\mathrm{MR}} = 0.015$ with a half-life near 692 days, broadly speaking.

3 Statistical Pair Screening and Regime Gating

3.1 Purpose and Building Blocks

Only, a pair should likely receive capital only if its spread has a tradeable mean-reverting component. Three statistics computed on the formation window (2,014 days, 2009-09-01 to 2017-08-30) carry the screen:

- R^2_{MR}, the share of mean-reverting variance (Proposition 1.4), computed with the difference-based formula (2) because R_t is varying;
- the half-life $\mathrm{HL} = -\ln 2/\ln \rho$ (3), where an optimiser that hits the bound $\rho = 0.999$ reports $\mathrm{HL} = 692.8$ days (marked * below; for these pairs a static fit halfly lives roughly 15–90 days according to the project notes);
- the ADF p-value, computed either on the raw static spread $X_{2,t} - \beta_0 X_{1,t}$ or on the *filtered* mean-reverting state $\hat{M}_t$ (Section 3.5), in many respects.

The screening question is a classification problem, namely anticipate, from formation data only, whether a pair will (in practical terms) be profitable out of sample, on the whole; each screen is therefore assessed as a classifier, with "WIN" defined as a positive out-of-sample total return of Static-PCI, to a significant degree.

3.2 Three Screening Frameworks

Framework 1: legacy strict screen. $R^2_{\mathrm{MR}} \geq 15\%$, $\mathrm{HL} \leq 252$ days, raw $p_{\mathrm{ADF}} \leq 0.10$, $\rho < 0.999$. It tests for *full* cointegration, but under PCI the raw spread contains the integrated term ε^R_t and a drifting β_t also affects a static spread, so the ADF test has low power. It rejects MRK–VZ, XOM–CVX and XLE–OIH, which later made money (Type II errors).

Framework 2: uncalibrated PCI screen. $R^2_{\mathrm{MR}} \geq 10\%$, $\mathrm{HL} \leq 365$ days, $p_{\mathrm{ADF}}(\hat{M}_t) \leq 0.05$. The filtered state is stationary almost by construction (e.g. $p \leq 10^{-8}$ for AAP–IBM and USO–XLE) even when R^2_{MR} is only 14–22%, i.e. most of the spread variance is drift. It admits such pairs (Static-PCI maximum drawdowns of -44.5% on AAP–IBM and -44.1% on USO–XLE), giving Type I errors.

Framework 3: calibrated variance-dominant PCI screen. Require $R^2_{\mathrm{MR}} \geq 50\%$ (the transient component carries most of the variance) and either raw $p_{\mathrm{ADF}} \leq 0.05$ or $p_{\mathrm{ADF}}(\hat{M}_t) \leq 0.01$.

3.3 Screening Metrics and Out-of-Sample Outcomes

Table 1 combines the formation diagnostics with out-of-sample outcomes for the 14 benchmark pairs, so as well, Figure 1 plots the diagnostics against the legacy thresholds, all things considered.

3.4 Classification Accuracy

The legacy screen perfectly precises (2/2) and specificity (8/8) but recall of only 33%; Framework 2 is no better than chance (five of its eight passes lost money); Framework 3 has precision 83%, recall 83% and specificity 88%, in many respects; in fact, its extra winners are MRK–VZ, XOM–CVX and XLE–OIH; HD–LOW is its only false positive and GS–CVX its only false negative, in the main.

Pair	R^2_{MR}	HL (d)	Raw p	$p(\hat{M}_t)$	Static ret.	Neural ret.	OOS	Legacy	Calib.
AAP–IBM	22.4%	22.0	0.3369	2×10^{-8}	−7.27%	−30.73%	L	R ✓	R ✓
EWA–EWC	78.6%	136.4	0.0439	0.0521	+1.89%	−2.90%	W	P ✓	P ✓
NKE–AES	21.6%	5.9	0.6494	10^{-17}	−64.40%	+45.70%	L	R ✓	R ✓
MRK–VZ	51.7%	692.8*	0.4882	4×10^{-4}	+37.72%	+40.23%	W	R ×	P ✓
GS–CVX	10.8%	498.5	0.1736	0.3103	+19.56%	+32.39%	W	R ×	R ×
KO–PEP	36.6%	692.8*	0.9641	8×10^{-16}	−9.11%	+3.92%	L	R ✓	R ✓
XOM–CVX	80.4%	692.8*	0.2432	0.0013	+20.54%	+40.11%	W	R ×	P ✓
GLD–SLV	41.0%	692.8*	0.3142	0.1266	−25.28%	−27.54%	L	R ✓	R ✓
XLE–XOP	13.4%	692.8*	0.9383	3×10^{-14}	−14.52%	−6.12%	L	R ✓	R ✓
XLE–OIH	62.5%	81.5	0.6900	0.0027	+15.57%	−5.76%	W	R ×	P ✓
USO–XLE	14.2%	4.2	0.3329	9×10^{-25}	−23.31%	−15.46%	L	R ✓	R ✓
MS–GS	90.8%	68.6	0.0491	5×10^{-4}	+16.37%	−11.65%	W	P ✓	P ✓
V–MA	33.9%	0.97	0.7121	$< 10^{-8}$	−8.93%	−14.04%	L	R ✓	R ✓
HD–LOW	50.9%	2.00	0.9874	3×10^{-25}	−21.42%	+6.83%	L	R ✓	P ×

Table 1: Formation-window screening metrics and out-of-sample outcomes. OOS: W/L = Static-PCI total return positive/negative. Legacy and Calib.: P = pass, R = reject; ✓/× = prediction correct/incorrect with respect to the OOS label.

Framework	Correct	TP	TN	FP	FN	Fisher exact p
1. Legacy strict	10/14 (71.4%)	2	8	0	4	0.165
2. Uncalibrated PCI	6/14 (42.9%)	3	3	5	3	1.000
3. Calibrated variance-dominant PCI	12/14 (85.7%)	5	7	1	1	0.026

Table 2: Screen outcomes against the sign of Static-PCI out-of-sample return (TP: passed and profitable; TN: rejected and unprofitable; FP: passed but lost money; FN: rejected but profitable). The last column is a two-sided Fisher exact test of association between pass/reject and win/loss.

Caveats. (i) The thresholds of Framework 3 were set *after* the out-of-sample outcomes were known, so 85.7% describes how well the rule separates these 14 pairs rather than predicting new ones, and the Fisher p-value is optimistic; walk-forward validation is needed. (ii) The WIN/LOSS label refers to Static-PCI only; the learned-gain model is profitable on NKE–AES and KO–PEP, which both screens reject. (iii) With 14 pairs, one pair moves accuracy by 7 points.

3.5 Online Regime Gate

Idea and inspiration. A pre-trade screen is evaluated once, but a pair can break *after* formation. The gate is a circuit breaker built from two cheap causal symptoms of a broken relationship: the spread becomes much more volatile than its own history, or it stops reverting (the estimated reversion speed falls or turns negative). If either appears, the position is flattened and entries are blocked. The idea draws on regime-switching models [26] (here replaced by deterministic detectors), rolling structural-break diagnostics [27], and mean-reversion-speed filters from statistical arbitrage practice [28, 29].

Definition. At each bar the gate is

$$g_t^{\mathrm{vol}} = \mathbb{1}\left[\frac{\mathrm{std}(\hat{M}_{t-20:t})}{\mathrm{std}(\hat{M}_{1:t})} \le 3\right], \tag{14}$$

$$g_t^{\mathrm{AR}} = \mathbb{1}[\kappa_t > 0 \text{ and } \mathrm{HL}_t^{\mathrm{roll}} \le 126], \quad \mathrm{HL}_t^{\mathrm{roll}} = \tfrac{\ln 2}{\kappa_t}, \tag{15}$$

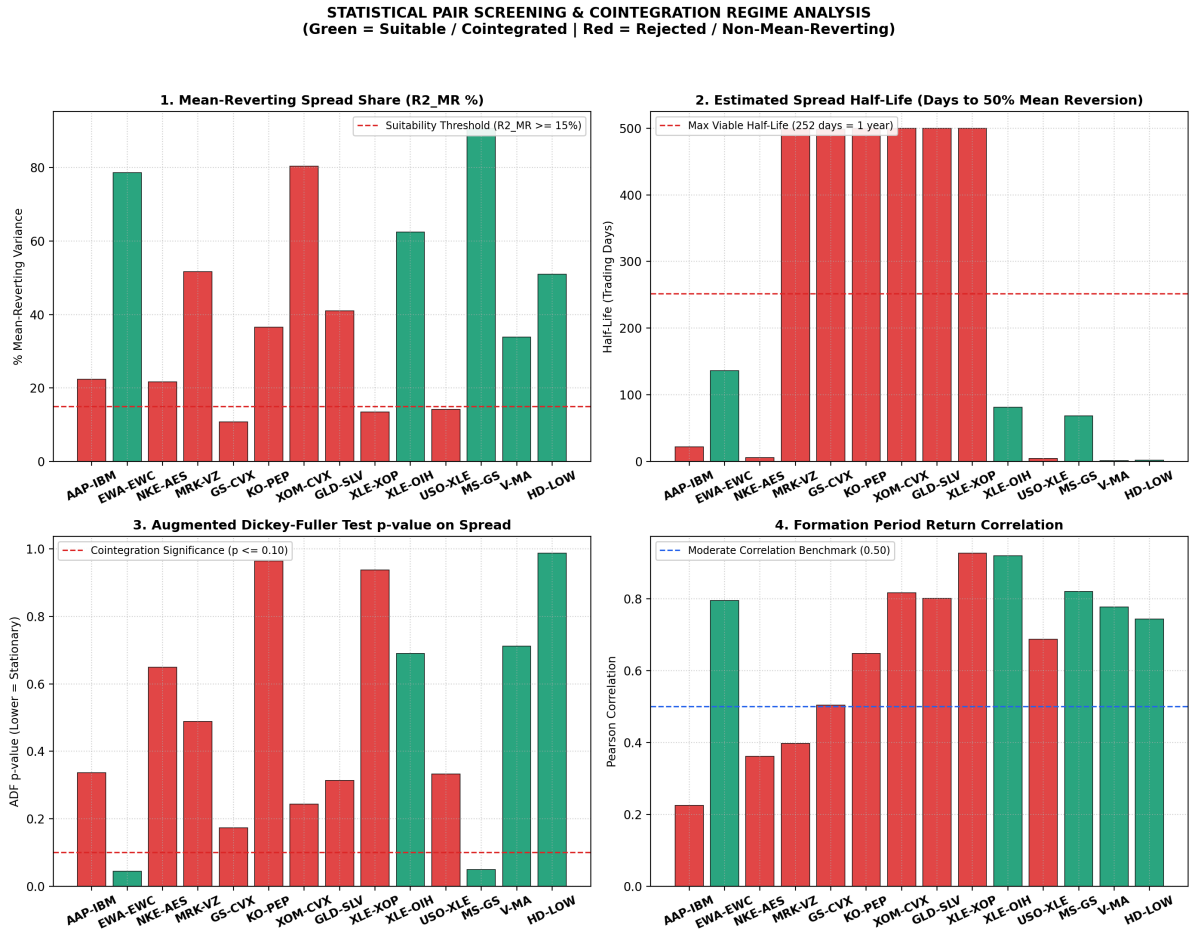


Figure 1: Screening diagnostics for the 14 benchmark pairs against the legacy (Framework 1) thresholds: mean-reverting variance share, half-life (capped at 500 for display), ADF p-value on the static spread and formation-period return correlation. Green: criterion passed; red: failed.

Over a 60-day window, where κ_t is estimated by OLS on the discrete Ornstein–Uhlenbeck regression $\Delta\hat{M}_s = \kappa(\mu - \hat{M}_{s-1}) + \epsilon_s$. If $g_t = 0$ an open position is liquidated and entries are blocked. And the composite gate is $g_t = g_t^{\text{vol}} g_t^{\text{AR}}$. The trade-off is between whipsaw (a sensitive gate exits profitable trades) and losses (a lax gate admits them). In the benchmark the Gated model created no positive return on any pair; in turn, even so, it lowered maximum drawdown relative to Dynamic-PCI on 9 of 14 (Section 8).

4 Neural KalmanBOT: Architecture

4.1 Design Rationale

Equations (5)–(8) show that in a linear model the gain $\boldsymbol{K}_t = \boldsymbol{P}_{t|t-1}\boldsymbol{H}_t^\top \boldsymbol{S}_t^{-1}$ rests on the unknown covariances $(\boldsymbol{Q}, \boldsymbol{R}_{\text{obs}})$ and on the recursively spread uncertainty $\boldsymbol{P}$, in real-world scenarios. When you try it (i) returns are heavy-tailed, so Gaussian-derived gains overreact to outliers and underreact to regime changes, (ii) $(\boldsymbol{Q}, \boldsymbol{R}_{\text{obs}})$ are estimated with discrepancy, and (iii) the model is mis-specified, e.g. β_t isn't an exact random walk, to a significant degree. KalmanNet [4] tackles all three by keeping the prediction step and replacing $\boldsymbol{K}_t$ by a recurrent network $\boldsymbol{K}_t^{\text{NN}}$; the network's hidden state plays the function of the covariance recursion. The overall framework therefore is remaining a filter (interpretable states $\hat{\beta}_t, \hat{M}_t, \hat{R}_t$) and doesn't need unambiguous $\boldsymbol{P}$

or matrix inversion, which also is decreasing quantitative issues.

Figure 2 shows one time step of the full framework and Figure 3 the internal network.

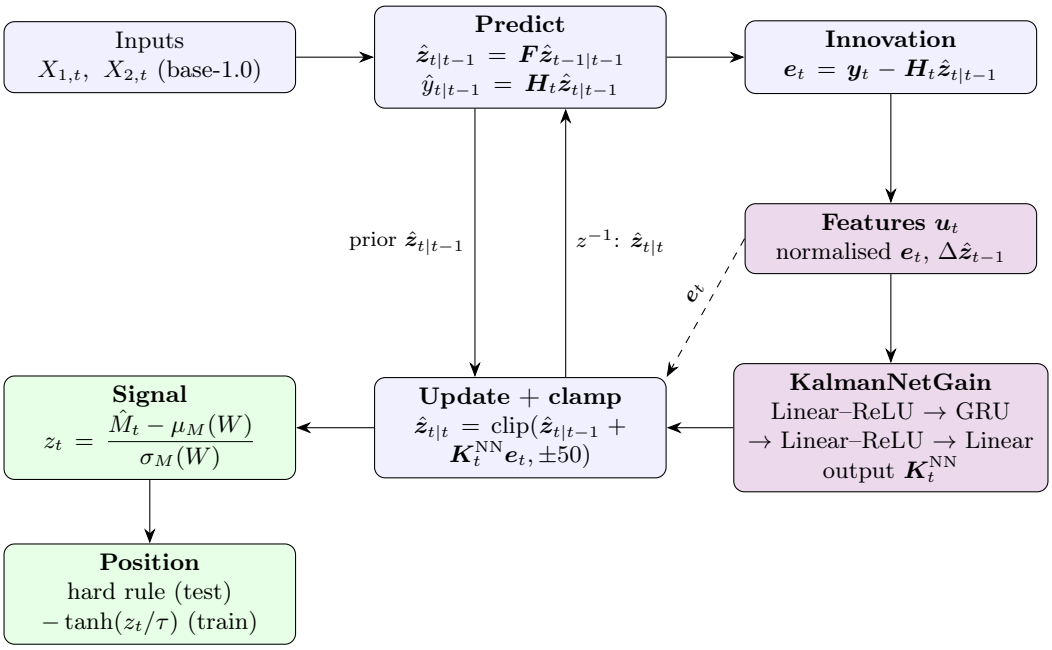


Figure 2: Computation at a single time step. The shaded violet blocks are learned; all other blocks are fixed model-based operations. The recurrent hidden state $\boldsymbol{h}_t$ of the GRU is carried across time (Figure 3); $\hat{\boldsymbol{z}}_{t|t}$ feeds the next prediction step through the unit delay z^{-1}.

4.2 Relation to Prior Work

The model-based prediction step of the Kalman filter is kept by KalmanNet [4] and learns the gain with a recurrent network (model-based deep learning [7]), at least empirically. Milstein et al. [2] (expanding [3]) are using it to pairs trading (KBPT) with a Bollinger-band policy on the z-score of the innovation (80-sample window) and two-step training, namely task-ignorant pre-training, then profit maximisation with a Gaussian-CDF replacement of the unit step, broadly speaking. Table 3 summarises the differences.

	Clegg & Krauss [1]	Milstein et al. [2]	This work
Spread model	AR(1)+RW, static β	AR(1) spread, dynamic h_t, μ_t	AR(1)+RW, dynamic β_t
Gain	analytical	GRU (KalmanNet)	GRU (KalmanNet)
Trading signal	z-score of $\hat{M}_t$	z-score of innovation Δy_t	z-score of filtered $\hat{M}_t$
Training objective	MLE	MSE → PnL (Gaussian-CDF surrogate)	MSE → Sharpe (tanh surrogate, cost-penalised)
Screening	variance ratio	pairs chosen a priori	variance-dominant PCI screen + online gate
Costs	–	omitted	10 bps per half-turn, in the loss

Table 3: Positioning of this project relative to the two main sources.

4.3 Input Features

The network receives two scale-invariant feature vectors. With m the state dimension and n the observation dimension,

$$\tilde{\boldsymbol{e}}_t = \frac{\boldsymbol{e}_t}{\|\boldsymbol{e}_t\|_2 + \epsilon} \in \mathbb{R}^n, \qquad \Delta\tilde{\boldsymbol{z}}_{t-1} = \frac{\hat{\boldsymbol{z}}_{t-1|t-1} - \hat{\boldsymbol{z}}_{t-1|t-2}}{\|\hat{\boldsymbol{z}}_{t-1|t-1} - \hat{\boldsymbol{z}}_{t-1|t-2}\|_2 + \epsilon} \in \mathbb{R}^m, \tag{16}$$

concatenated as $\boldsymbol{u}_t = [\tilde{e}_t; \Delta\tilde{\boldsymbol{z}}_{t-1}] \in \mathbb{R}^{m+n}$; these are corresponding to features F2 (innovation) and F4 (forward update difference) of KalmanNet [4], all things considered.

Remark 4.1 (Two corrections to a naive feature set). (i) One might perhaps encompass $\hat{\boldsymbol{z}}_{t|t-1} - \boldsymbol{F}\hat{\boldsymbol{z}}_{t-1|t-1}$, but this is identically zero by (5); the informative forward-update difference is therefore the *previous* step's correction $\hat{\boldsymbol{z}}_{t-1|t-1} - \hat{\boldsymbol{z}}_{t-1|t-2} = \boldsymbol{K}_{t-1}\boldsymbol{e}_{t-1}$, at least in practice. (ii) When $n = 1$, normalising a scalar innovation gives $e_t/(|e_t| + \epsilon) \approx \operatorname{sign}(e_t)$, discarding size; the production network therefore also exposes the two-dimensional observation $\boldsymbol{y}_t = [X_{2,t}, X_{1,t}]^\top$ and the prior state $\hat{\boldsymbol{z}}_{t|t-1}$, which preserves relative scale; this enlarges only the first linear layer, to a significant degree.

4.4 Layer-by-Layer Specification

The network (`KalmanNetGain`) has four trainable stages (Figure 3):

1. **Input projection:** $\boldsymbol{h}_t^{(1)} = \mathrm{ReLU}(\boldsymbol{W}_1\boldsymbol{u}_t + \boldsymbol{b}_1) \in \mathbb{R}^{H_1}$, $H_1 = \min(48, (m+n)\cdot 8)$: 32 for the production 3-state model (48 for the legacy 4-state model), in many respects.
2. **Recurrent core:** a single-layer GRU with hidden size $d_h = 32$,

$$\boldsymbol{r}_t = \sigma(\boldsymbol{W}_{ir}\boldsymbol{h}_t^{(1)} + \boldsymbol{W}_{hr}\boldsymbol{h}_{t-1} + \boldsymbol{b}_r), \qquad \boldsymbol{g}_t = \sigma(\boldsymbol{W}_{ig}\boldsymbol{h}_t^{(1)} + \boldsymbol{W}_{hg}\boldsymbol{h}_{t-1} + \boldsymbol{b}_g), \tag{17}$$

$$\boldsymbol{n}_t = \tanh\big(\boldsymbol{W}_{in}\boldsymbol{h}_t^{(1)} + \boldsymbol{b}_{in} + \boldsymbol{r}_t \odot (\boldsymbol{W}_{hn}\boldsymbol{h}_{t-1} + \boldsymbol{b}_{hn})\big), \quad \boldsymbol{h}_t = (1 - \boldsymbol{g}_t)\odot\boldsymbol{n}_t + \boldsymbol{g}_t \odot \boldsymbol{h}_{t-1}. \tag{18}$$

   The reset gate $\boldsymbol{r}_t$ and update gate $\boldsymbol{g}_t$ [19] let the network decide how much past innovation history to keep; $\boldsymbol{h}_t$ acts as a learned summary of the running covariance information that the analytical filter carries in $\boldsymbol{P}_{t|t-1}$, in real-world scenarios.
3. **Hidden projection:** $\boldsymbol{h}_t^{(2)} = \mathrm{ReLU}(\boldsymbol{W}_2\boldsymbol{h}_t + \boldsymbol{b}_2) \in \mathbb{R}^{H_2}$, $H_2 = \min(24, \max(8, 2mn))$.
4. **Gain head:** $\boldsymbol{k}_t = \boldsymbol{W}_{\text{out}}\boldsymbol{h}_t^{(2)} + \boldsymbol{b}_{\text{out}}$, reshaped to $\boldsymbol{K}_t^{\text{NN}} \in \mathbb{R}^{m\times n}$. The weights and bias are, after all, initialised to *zero*.

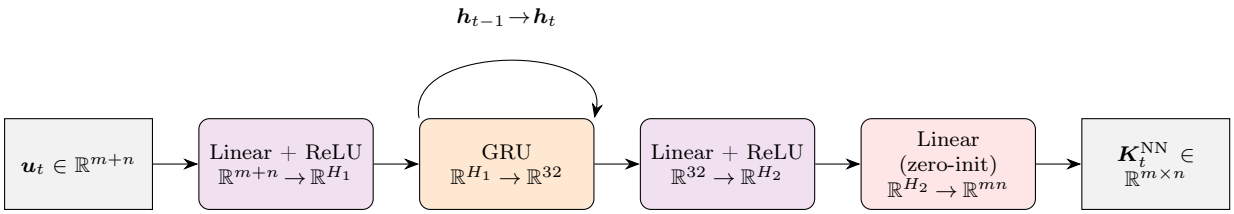


Figure 3: Internal structure of the learned gain network. Only the final linear head is zero-initialised; the GRU maintains temporal memory of innovations.

Parameter counts. Counting weights and biases directly from the layer definitions (PyTorch GRU has two bias vectors per gate, giving $3\,[H_1 d_h + d_h^2 + 2d_h]$ parameters) yields Table 4; the network is deliberately small. For reference, KalmanNet [4] reports of the order of 5×10^5 trainable coefficients for its first architecture and 2.5×10^4 for its second; the network here's smaller, which suits the roughly 2,000 formation observations per pair, in the main.

Layer	3-state, production (m=3, n=1)	4-state, legacy (m=4, n=2)
Input Linear ($m+n \to H_1$)	$4 \cdot 32 + 32 = 160$	$6 \cdot 48 + 48 = 336$
GRU ($H_1 \to 32$)	$3(32 \cdot 32 + 32^2 + 64) = 6{,}336$	$3(48 \cdot 32 + 32^2 + 64) = 7{,}872$
Hidden Linear ($32 \to H_2$)	$32 \cdot 8 + 8 = 264$	$32 \cdot 16 + 16 = 528$
Gain head ($H_2 \to mn$)	$8 \cdot 3 + 3 = 27$	$16 \cdot 8 + 8 = 136$
Total	**6,787**	**8,872**

Table 4: Trainable parameters of `KalmanNetGain` for the production 3-state model and the discarded 4-state model (canonical $(m+n)$-dimensional input).

Why zero initialisation? With $\boldsymbol{W}_{\text{out}} = \mathbf{0}, \boldsymbol{b}_{\text{out}} = \mathbf{0}$, $\boldsymbol{K}_t^{\text{NN}} \equiv \mathbf{0}$ initially, so $\hat{\boldsymbol{z}}_{t|t} = \hat{\boldsymbol{z}}_{t|t-1}$ and the filter starts by trusting its prior. Earlier layers receive no gradient on the first step (it is multiplied by $\boldsymbol{W}_{\text{out}}^{\top} = \mathbf{0}$), but $\boldsymbol{W}_{\text{out}}$ and $\boldsymbol{b}_{\text{out}}$ do, after which all layers train.

Posterior update with clamping.

$$\hat{\boldsymbol{z}}_{t|t} = \text{clip}\big(\hat{\boldsymbol{z}}_{t|t-1} + \boldsymbol{K}_t^{\text{NN}} \boldsymbol{e}_t,\, -50,\, 50\big). \tag{19}$$

Keep in mind that the network learns the whole gain matrix, so it can (at least in most settings) allocate the innovation, across (β_t, M_t, R_t) differently from the analytical filter, at least in principle; no covariance $\boldsymbol{P}_t$ is spread in the neural model.

5 Training Algorithm

5.1 Two-Stage Curriculum

The formation (training) split is being employed by training, and no ground-truth values of (β_t, M_t, R_t) exist for real data, so the supervised KalmanNet loss on states is unavailable, all things considered. We follow the unsupervised idea of [5, 2]: train on the observation prediction error, then fine-tune on the trading objective, to a significant degree.

5.2 Stage 1: Observation-MSE Pre-training

$\mathcal{L}_{\text{MSE}}$ (Figure 4) is the one-step-ahead prediction error of the follower; under the Gaussian model, minimising it's (up to the $\ln|\boldsymbol{S}_t|$ term) the Gaussian likelihood contribution, so the gain is first driven toward the values that are making innovations small, i.e. toward the analytical filter, on the whole. Milstein et al. [2] report that training on profit alone, without this warm start, often branches off.

5.3 Stage 2: Differentiable Sharpe-Ratio Fine-Tuning

Hard trading thresholds have no gradient, so, as in [2] (Gaussian-CDF surrogate), a smooth position proxy is usually employed:

$$\text{pos}_t^g = -\tanh\left(\frac{z(\hat{M}_t)}{\tau_{\text{soft}}}\right), \qquad \tau_{\text{soft}} = 1, \tag{20}$$

$$R_t^{\text{strat}} = \text{pos}_{t-1}^g \left(w_{2,t-1}\frac{\Delta X_{2,t}}{X_{2,t-1}} - w_{1,t-1}\frac{\Delta X_{1,t}}{X_{1,t-1}}\right) - c\,|\text{pos}_t^g - \text{pos}_{t-1}^g|, \tag{21}$$

$$\mathcal{L}_{\text{Sharpe}} = -\sqrt{252}\,\frac{\mathbb{E}[R_t^{\text{strat}}]}{\sqrt{\text{Var}(R_t^{\text{strat}}) + 10^{-8}}}. \tag{22}$$

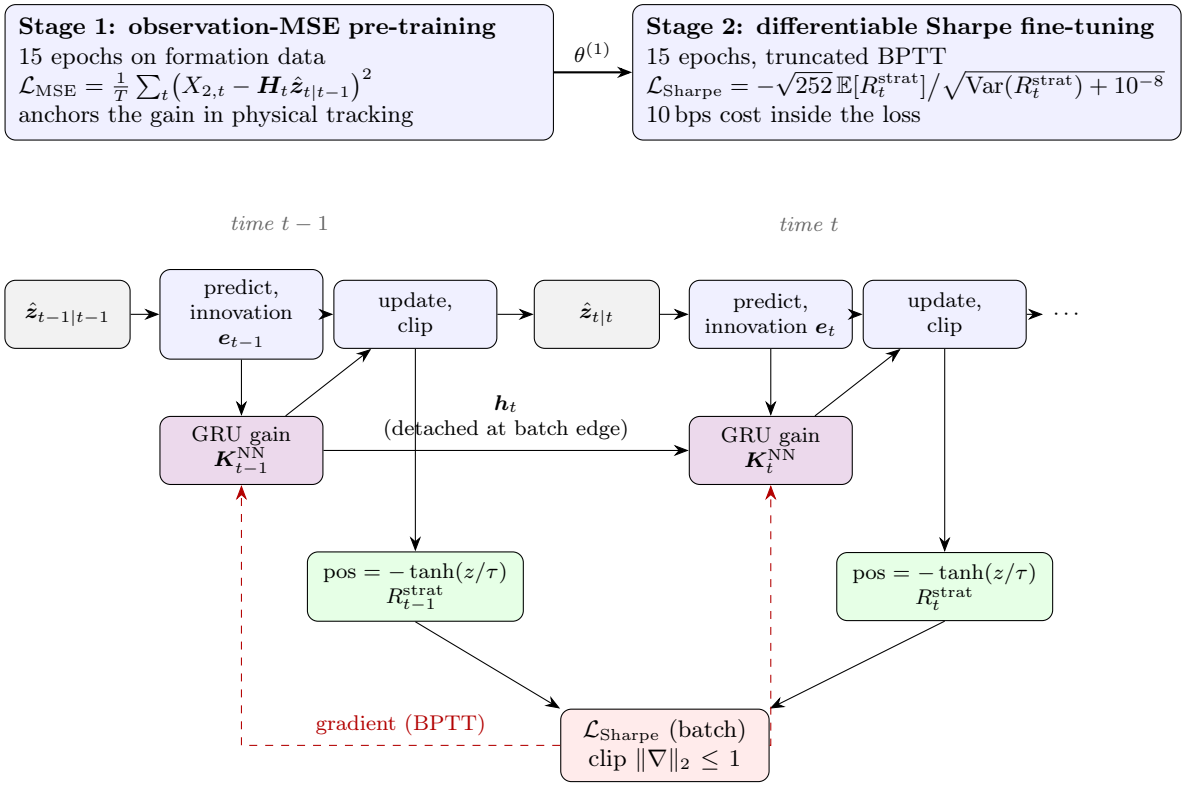


Figure 4: Two-stage training. Top: the curriculum. Bottom: the unrolled computation graph used in Stage 2 (two time steps shown). Violet blocks are learned; dashed red arrows show gradients flowing from the Sharpe loss back through the position proxy, the clipped update and the GRU across time. The recurrent hidden state is detached across batch boundaries.

A large positive z_t (spread rich) poses. Seeing as $\tanh(x) = 2\,\text{sigm}(2x) - 1$, (20) is a logistic-CDF surrogate of the same family as the Gaussian CDF; Figure 5 is comparing it with the hard rule, in real-world scenarios.

Proposition 5.1 (Gradient of the Sharpe loss)**.** *Let* $\mu = \frac{1}{T}\sum_t R_t$, $\sigma^2 = \frac{1}{T}\sum_t (R_t - \mu)^2$ *and* $\mathcal{L} = -\sqrt{252}\,\mu/\sigma$. *Then*

$$\frac{\partial \mathcal{L}}{\partial R_t} = -\frac{\sqrt{252}}{T\sigma}\Big[1 - \frac{\mu}{\sigma}\cdot\frac{R_t - \mu}{\sigma}\Big].$$

Proof. $\partial\mu/\partial R_t = 1/T$ and $\partial\sigma/\partial R_t = (R_t - \mu)/(T\sigma)$. Employing the quotient rule to μ/σ gives $\frac{1}{T\sigma} - \frac{\mu(R_t-\mu)}{T\sigma^3}$. □

Each day's return is typically weighted by $1 - \text{SR}_d(R_t - \mu)/\sigma$ (SR_d the daily Sharpe), so unusually large gains earn less slight credit, on the whole. What's more, the loss favours consistency over high-variance bets, so the chain rule then is continuing through $R_t = \text{pos}_{t-1}^g r_t - c|\Delta\text{pos}|$, $\partial\text{pos}/\partial z = -\tau^{-1}\text{sech}^2(z/\tau)$ (which vanishes for $|z| \gg \tau$) and $\hat{M}_t \mapsto \boldsymbol{K}_t^{\text{NN}}$, at least in principle. The cost term penalises turnover directly. The loss is generally computed on training data, so it's an in-sample method.

5.4 Numerical Stabilisation

Unrolling the filter gives $\partial\hat{\boldsymbol{z}}_{T|T}/\partial\hat{\boldsymbol{z}}_{0|0} \approx \prod_t (\boldsymbol{I} - \boldsymbol{K}_t^{\text{NN}}\boldsymbol{H}_t)\boldsymbol{F}$ (ignoring the state-dependence of $\boldsymbol{K}_t^{\text{NN}}$), which can grow fast if the spectral radius exceeds 1; $\boldsymbol{H}_t$ comprises $X_{1,t}$, so large price

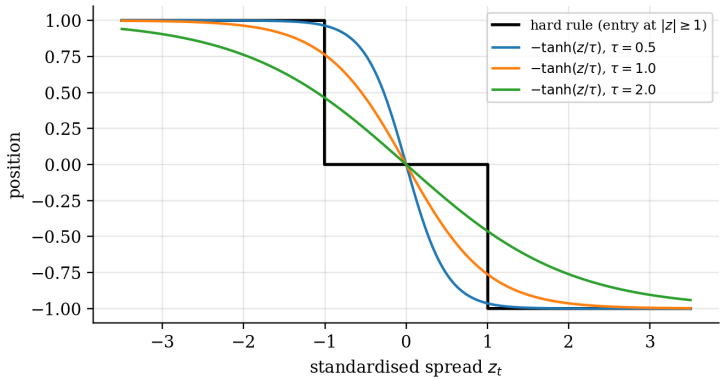


Figure 5: The smooth position proxy used during training versus the hard entry rule. Smaller τ approximates the hard rule more closely but has vanishing gradients in the tails; larger τ gives informative gradients but a poorer match to trading.

levels inflate $\boldsymbol{KH}$. An unconstrained gain can therefore feed back on itself and, with the tanh derivative, overflow to `NaN`. The fixes were:

1. **Hard state clamping** (19), bounding the recursion;
2. **Capacity regularisation**: GRU hidden size reduced from 200 to 32 units;
3. **Gradient-norm clipping** $\|\boldsymbol{g}\|_2 \leq 1$, the standard fix for exploding gradients in recurrent networks [21];
4. **Hidden-state detaching** across the batch boundaries ($\boldsymbol{h}_{t-1} \leftarrow \boldsymbol{h}_{t-1}$.detach()), i.e. truncated BPTT, which bounds the length of the product above;
5. **Base-1.0 price normalisation**, so that $\boldsymbol{H}_t$ entries are $O(1)$, together with the zero-initialised gain head.

Clamping and zero-initialisation are usually heuristics; they don't guarantee Lyapunov stability of the learned filter, in the main.

Training configuration. Table 5 lists the settings under which the stabilised training was run.

Setting	Value
Stage 1 epochs / Stage 2 epochs	15 / 15 (5 each in the `-quick` test mode)
GRU hidden size; H_1; H_2	32; $\min(48, 8(m{+}n))$; $\min(24, \max(8, 2mn))$
Soft-position temperature τ_{soft}	1.0
Transaction cost in loss c	0.0010 (10 bps)
Annualisation / variance floor	$\sqrt{252}$ / 10^{-8}
State clamp	$[-50, 50]$
Gradient-norm clip	1.0
Training data	formation window only (2,014 trading days)
Optimiser, learning rate, seed	as set in `neural_kalman_pci.py` (see Section 11.2)

Table 5: Training configuration of Neural KalmanBOT.

6 Trading Strategy and Execution

6.1 Signal Construction and Rules

Signals are created only from the filtered temporary state, never from R_t:

$$z_t = \frac{\hat{M}_{t|t} - \mu_M(W)}{\sigma_M(W)}, \qquad W = 80 \text{ days}, \tag{23}$$

Honestly, where $\mu_M(W)$ and $\sigma_M(W)$ are the rolling mean and standard departure of $\hat{M}$ (the window length is often taken from [2]). The rules are:

- **Long-spread entry:** $z_t \leq -\tau_{\text{open}}$ (and gate on): buy X_2, short $\beta_t X_1$. Production $\tau_{\text{open}} = 1.0$.
- **Short-spread entry:** $z_t \geq +\tau_{\text{open}}$: short X_2, buy $\beta_t X_1$.
- **Exit:** full close to cash when z_t crosses zero.
- **Stop-loss:** emergency liquidation at a position drawdown of -10%.

Employing $\hat{M}$ rather than the raw innovation (as in [2]) removes the lasting component from the signal, in the main; and thus, only the same, it depends on the filter's ability to separate M_t from R_t, which Proposition 1.5 is showing is only partial, at least in principle.

6.2 Dollar-Neutral Sizing and Net Returns

One asset is bought by a spread position and sells the other in proportion to the hedge ratio, after that there's another way to say it's it's. Capital is normally scaled so that the two legs together are using one unit of gross exposure, including the weights below sum (in absolute value) to one and their ratio is the hedge ratio, in actual application. Weights rebalance daily and a noisy $\hat{\beta}_t$ is raising turnover and costs, because β_t is typically re-estimated daily, at least in practice; in fact, the legs are equal in dollar terms only when $|\beta_t| = 1$, so "dollar-neutral" here means hedge-ratio-weighted legs with unit gross exposure. For $\beta_t = 1.5$, $w_2 = 0.40$ and $w_1 = 0.60$ (40 % long X_2, 60 % short X_1); for $\beta_t = 1$ both legs are generally 50 %.

Weights are, after all, figured out from the posterior hedge ratio,

$$w_{2,t-1} = \frac{1}{1 + |\beta_{t-1}|}, \qquad w_{1,t-1} = \frac{\beta_{t-1}}{1 + |\beta_{t-1}|}, \tag{24}$$

and the net daily return with proportional cost $c = 0.0010$ per half-turn on each leg is

$$r_t^{\text{pair}} = \text{pos}_{t-1}\left(w_{2,t-1}\frac{X_{2,t}-X_{2,t-1}}{X_{2,t-1}} - w_{1,t-1}\frac{X_{1,t}-X_{1,t-1}}{X_{1,t-1}}\right) - 2c\mathbb{1}(\text{pos}_t \neq \text{pos}_{t-1}), \quad E_t = \prod_{s\leq t}(1+r_s^{\text{pair}}). \tag{25}$$

The training loss (21) is employing the same cost scale so the learned gain is mainly penalised for turnover, all things considered; seeing as the dollar weights sum to one, each leg carries a fraction of capital that depends on β_t, in many respects.

6.3 The Four Models Compared

All four models share the same data, pre-trade screen, signal rule and cost model, so differences are currently attributable to the filter, at least for now. The comparisons are usually designed as ablations: **Static vs. Dynamic** separates the value of a time-varying hedge ratio; **Dynamic vs. Gated** separates the online gate; **Dynamic vs. Neural** is isolating the learned gain (same state model, same β_t dynamics), in real-world scenarios.

Feature	Static-PCI	Dynamic-PCI	Regime-Gated	Neural Kalman-BOT
Hedge ratio β_t	fixed β_0 (MLE)	random walk	random walk	random walk
Gain $\boldsymbol{K}_t$	analytical	analytical	analytical	GRU-learned
Spread model	AR(1)+RW	AR(1)+RW	AR(1)+RW	AR(1)+RW
Parameters fitted by	MLE, $q_\beta \approx 0$	MLE, 5 params	MLE, 5 params	MLE (prediction step) + network
Risk control	screen only	screen only	online dual gate	learned gain attenuation
Cost awareness	accounting only	accounting only	accounting only	inside the loss

Table 6: Architectural comparison of the four evaluated methods.

7 Synthetic Ground-Truth Validation

Before employing real data, the estimation machinery was checked on a imitated data-generating process (DGP) with known states, like a driver X_1 following a random walk from 100, $X_2 = \beta_t X_1 + M_t + R_t$ with $\beta_0 = 1.5$, $\rho = 0.9$, $\sigma_M = 0.5$, $\sigma_R = 0.3$, over $n = 1{,}250$ days, broadly speaking, and factors are estimated by L-BFGS-B on (9) and the filter is then run with the projections, to a significant degree.

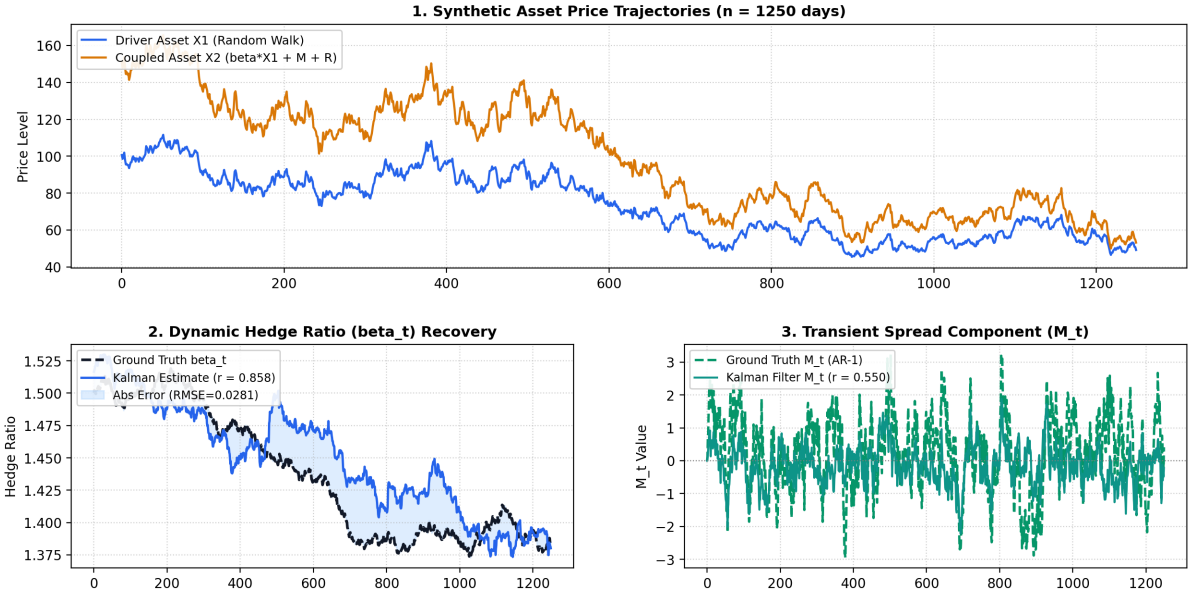


Figure 6: Synthetic validation (adapted from the project's `simulation_results` figure; panels 1–3). The dynamic hedge ratio is recovered with correlation $r = 0.858$ (RMSE 0.0281), but the filtered transient state reproduces the true M_t with correlation only $r = 0.550$ and visibly shrunken amplitude.

Interpretation. β_0 and ρ are recovered to about 1%, but σ_R is underestimated by 26% and σ_M by 6%; this is the weak identification of Proposition 1.5, although one simulated path cannot establish bias. The filtered $\hat{M}_t$ is a conditional mean and so has smaller variance than M_t ($\mathrm{Var}(M_t) = \mathrm{Var}(\hat{M}_t) + \mathbb{E}[(M_t - \hat{M}_t)^2]$); the visible shrinkage is expected, and the z-score (23) normalises by the rolling standard deviation of $\hat{M}$ itself.

Whitening test (re-simulated). The original validation figure had a mis-scaled panel for normalised innovations (a single spike and a flat reference curve), so it was replaced by an

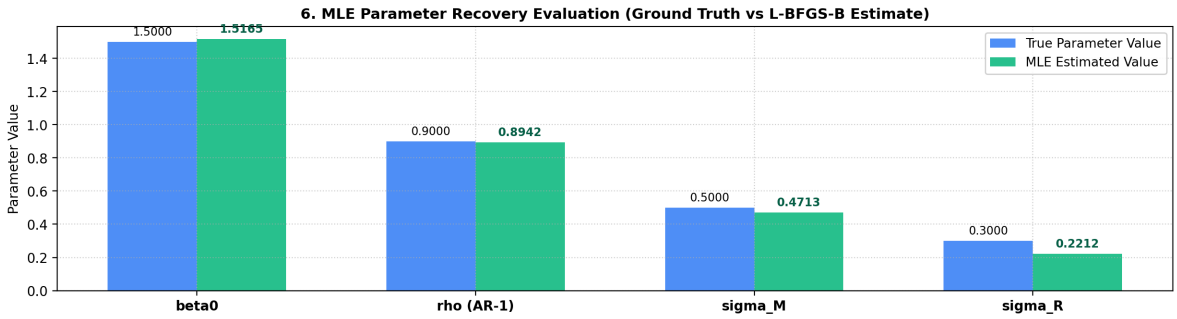


Figure 7: Maximum-likelihood parameter recovery (original panel 6). Relative errors: β_0 +1.1%, ρ −0.6%, σ_M −5.7%, σ_R −26.3%.

independent re-simulation of the same process ($q_\beta = 0.003$, $\sigma_{\text{obs}} = 0.1$, unit-variance X_1 steps, seed 7) with the filter run at the true parameters. Over 1,230 post-burn-in innovations the mean is 0.012, standard deviation 1.015, skewness −0.05 and excess kurtosis −0.03; the Ljung–Box test [25] (20 lags) gives $p = 0.52$, Kolmogorov–Smirnov $p = 0.69$ and Shapiro–Wilk $p = 0.75$ (Figure 8). This checks the filter implementation and likelihood, not model fit on real data.

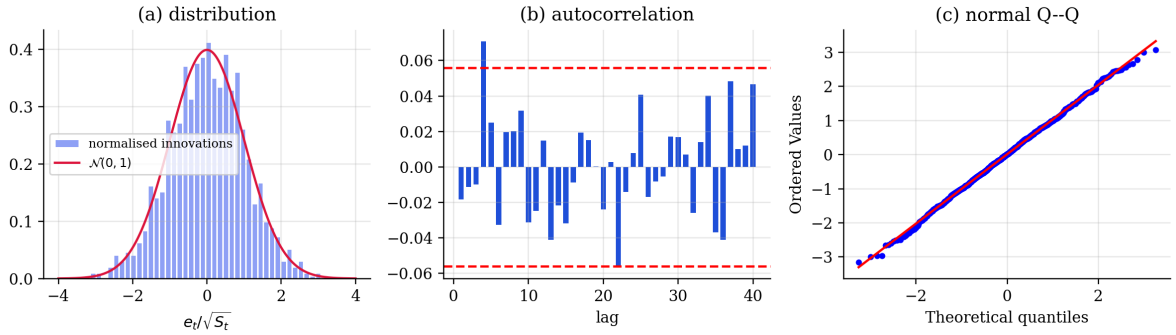


Figure 8: Whitening test on an independent re-simulation (true parameters). (a) normalised innovations vs. $\mathcal{N}(0,1)$; (b) autocorrelation with 95% bands; (c) normal Q–Q plot.

8 Empirical Benchmark and Results

8.1 Data and Protocol

The universe includes 14 pairs across the single-stock equities, sovereign and sector ETFs and commodity/energy proxies, such as AAP–IBM, EWA–EWC, NKE–AES, MRK–VZ, GS–CVX, KO–PEP, XOM–CVX, GLD–SLV, XLE–XOP, XLE–OIH, USO–XLE, MS–GS, V–MA and HD–LOW, on the whole, and daily prices come from Yahoo Finance and are currently normalised to base 1.0.

- **Formation (in-sample):** 2,014 trading days, 2009-09-01 to 2017-08-30: all parameter estimation and neural training.
- **Trading (out-of-sample):** 1,007 trading days, 2017-08-31 to 2021-08-31, spanning the COVID-19 crash and recovery.
- **Costs:** 10 bps per half-turn on both legs for every rebalance (all models).

Eleven of the pairs were chosen as a varied cross-asset set and three (MS–GS, V–MA, HD–LOW) were added after the screening study of Section 3, at least empirically; indeed, sure enough, six of the 14 pairs pass the calibrated screen (Table 7); the remainder are deliberately included as

stress tests of robustness, all things considered.

8.2 Pair-Level Results

Table 7 lists total return, annualised return, Sharpe ratio and maximum drawdown for every model and pair, at least in practice. Figures 9–11 are displaying the same data graphically and Figure 12 shows all equity curves.

Table 7: Out-of-sample performance, 1,007 trading days, 10 bps costs. Screen, namely P = passes the adjusted variance-dominant screen of Section 5, R = rejected, to a significant degree; bold: highest Sharpe within a pair.

Pair	Screen	Model	Total return	Ann. return	Sharpe	Max DD
AAP–IBM	R	Static-PCI	−7.27%	−1.87%	**0.01**	−44.52%
		Dynamic-PCI	−24.59%	−6.82%	−0.27	−43.55%
		Regime-Gated	−28.72%	−8.12%	−0.38	−45.44%
		Neural KalmanBOT	−30.73%	−8.78%	−0.41	−47.23%
EWA–EWC	P	Static-PCI	+1.89%	+0.47%	**0.11**	−9.30%
		Dynamic-PCI	−5.16%	−1.32%	−0.18	−13.21%
		Regime-Gated	−3.69%	−0.94%	−0.16	−14.31%
		Neural KalmanBOT	−2.90%	−0.73%	−0.08	−14.45%
NKE–AES	R	Static-PCI	−64.40%	−22.78%	−0.69	−69.47%
		Dynamic-PCI	−15.97%	−4.26%	−0.06	−52.64%
		Regime-Gated	−10.34%	−2.69%	−0.01	−46.71%
		Neural KalmanBOT	+45.70%	+9.88%	**0.55**	−23.04%
MRK–VZ	P	Static-PCI	+37.72%	+8.34%	**0.82**	−7.00%
		Dynamic-PCI	−0.25%	−0.06%	0.05	−15.13%
		Regime-Gated	−5.74%	−1.47%	−0.18	−13.19%
		Neural KalmanBOT	+40.23%	+8.83%	0.75	−10.38%
GS–CVX	R	Static-PCI	+19.56%	+4.57%	0.40	−24.30%
		Dynamic-PCI	−1.80%	−0.45%	0.03	−26.48%
		Regime-Gated	−0.13%	−0.03%	0.04	−16.50%
		Neural KalmanBOT	+32.39%	+7.27%	**0.41**	−43.86%
KO–PEP	R	Static-PCI	−9.11%	−2.36%	−0.27	−23.18%
		Dynamic-PCI	−2.83%	−0.72%	−0.04	−14.33%
		Regime-Gated	−6.77%	−1.74%	−0.36	−11.51%
		Neural KalmanBOT	+3.92%	+0.97%	**0.15**	−10.18%
XOM–CVX	P	Static-PCI	+20.54%	+4.79%	0.59	−9.83%
		Dynamic-PCI	−11.99%	−3.15%	−0.23	−32.80%
		Regime-Gated	−4.07%	−1.03%	−0.16	−15.15%
		Neural KalmanBOT	+40.11%	+8.81%	**0.98**	−11.12%
GLD–SLV	R	Static-PCI	−25.28%	−7.03%	−1.11	−25.51%
		Dynamic-PCI	+13.24%	+3.16%	**0.46**	−16.75%
		Regime-Gated	−2.73%	−0.69%	−0.25	−6.48%
		Neural KalmanBOT	−27.54%	−7.75%	−1.26	−27.71%
XLE–XOP	R	Static-PCI	−14.52%	−3.85%	−0.43	−20.35%
		Dynamic-PCI	−10.56%	−2.75%	**−0.07**	−40.64%
		Regime-Gated	−0.58%	−0.15%	−0.68	−0.58%
		Neural KalmanBOT	−6.12%	−1.57%	−0.13	−16.91%
XLE–OIH	P	Static-PCI	+15.57%	+3.69%	**0.35**	−19.73%
		Dynamic-PCI	+16.47%	+3.89%	0.28	−40.06%
		Regime-Gated	−31.92%	−9.17%	−0.75	−39.27%

Pair	Screen	Model	Total return	Ann. return	Sharpe	Max DD
		Neural KalmanBOT	−5.76%	−1.47%	−0.12	−13.57%
USO–XLE	R	Static-PCI	−23.31%	−6.43%	−0.27	−44.09%
		Dynamic-PCI	−34.82%	−10.16%	−0.36	−56.53%
		Regime-Gated	−37.56%	−11.12%	−0.42	−56.53%
		Neural KalmanBOT	−15.46%	−4.12%	**−0.10**	−29.86%
MS–GS	P	Static-PCI	+16.37%	+3.87%	**0.50**	−9.37%
		Dynamic-PCI	−16.77%	−4.49%	−0.42	−22.93%
		Regime-Gated	−17.55%	−4.71%	−0.66	−23.65%
		Neural KalmanBOT	−11.65%	−3.05%	−0.22	−26.21%
V–MA	R	Static-PCI	−8.93%	−2.31%	**−0.46**	−12.33%
		Dynamic-PCI	−8.84%	−2.29%	−0.54	−14.30%
		Regime-Gated	−8.84%	−2.29%	−0.54	−14.30%
		Neural KalmanBOT	−14.04%	−3.72%	−0.69	−15.27%
HD–LOW	P	Static-PCI	−21.42%	−5.85%	−0.52	−26.74%
		Dynamic-PCI	−11.90%	−3.12%	−0.33	−20.05%
		Regime-Gated	−11.67%	−3.06%	−0.32	−19.84%
		Neural KalmanBOT	+6.83%	+1.67%	**0.23**	−18.56%

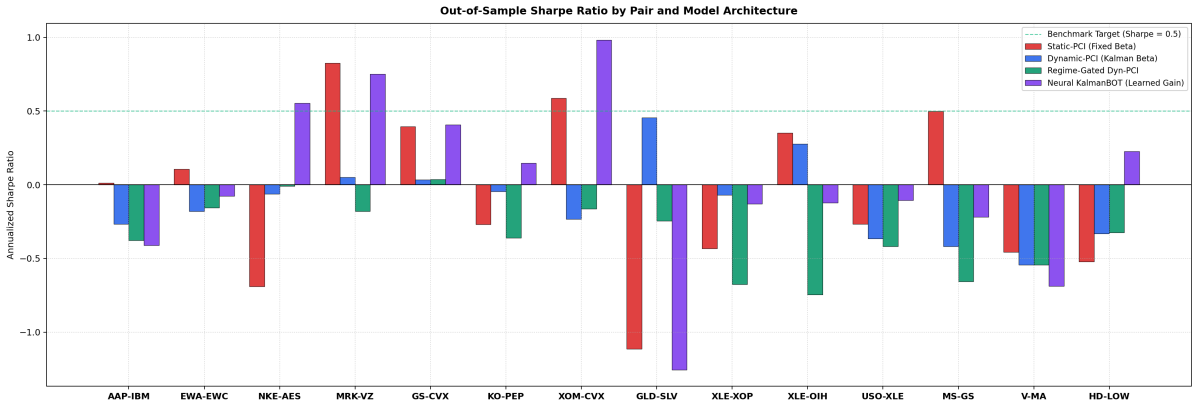


Figure 9: Out-of-sample annualised Sharpe ratio by pair and model. The dashed line marks the 0.5 reference level.

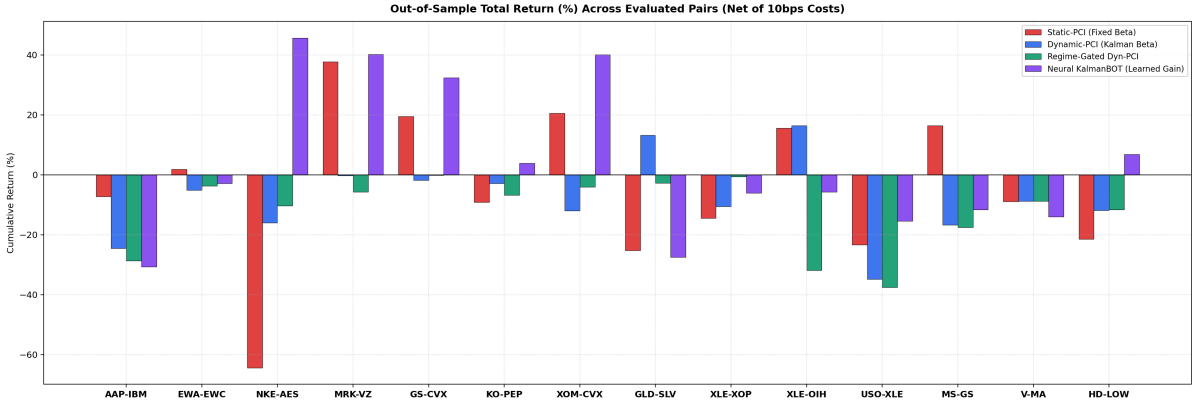


Figure 10: Out-of-sample total return (%) net of 10 bps costs.

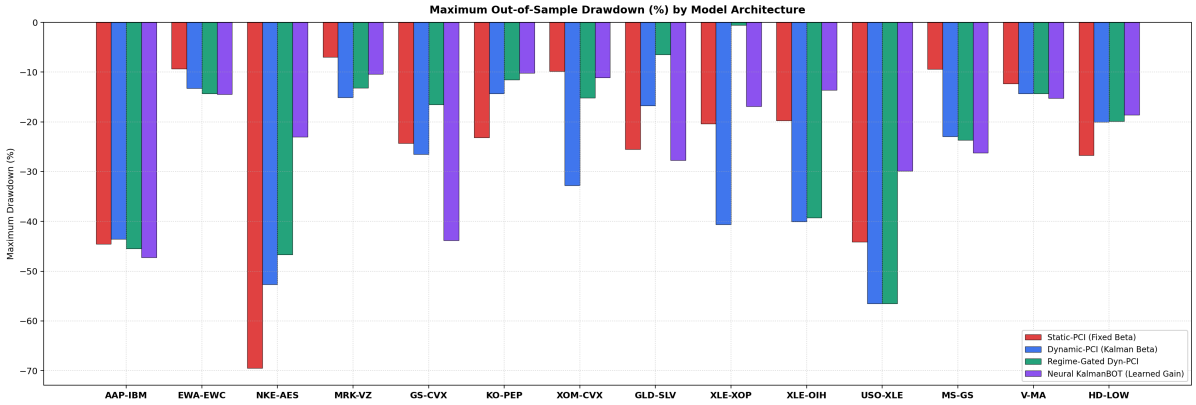


Figure 11: Maximum out-of-sample drawdown (%) by pair and model.

8.3 Aggregate Performance Across Pairs

Across the 14 pairs (QLATEXz0000q, because pair-level outcomes are, after all, mixed probably, each model is summarised; these summaries are computed from QLATEXz0001q, at least empirically. Neural KalmanBOT achieves the highest mean return, the smallest worst-case draw-

Model	Mean ret.	Median ret.	Mean Sharpe	# ret.>0	Mean MaxDD	Worst MaxDD	Mean ex-NKE
Static-PCI	−4.5%	−8.1%	−0.07	6/14	−24.7%	−69.5%	+0.1%
Dynamic-PCI	−8.3%	−9.7%	−0.12	2/14	−29.2%	−56.5%	−7.7%
Regime-Gated	−12.2%	−7.8%	−0.35	0/14	−23.1%	−56.5%	−12.3%
Neural KalmanBOT	+3.9%	−4.3%	0.00	6/14	−22.0%	−47.2%	+0.7%

Table 8: Cross-pair summary statistics. The last column excludes the single most extreme pair (NKE–AES).

down, and the smallest mean drawdown across the evaluated strategies. However, its *median* return remains negative, and it generates positive returns on only 6 of the 14 pairs, matching Static-PCI. Furthermore, excluding NKE–AES reduces its mean return to approximately +0.7%, only marginally different from the +0.1% mean return of Static-PCI. This suggests that Neural KalmanBOT's aggregate performance is disproportionately influenced by a small number of large winners, namely NKE–AES, MRK–VZ, GS–CVX, and XOM–CVX, each of which delivers a return exceeding +30%.

8.4 Findings by Pair Type

Grouping the pairs by outcome (post-hoc; screening outcomes are often in Section 3):

- **Static-PCI profitable (MRK–VZ, XOM–CVX, GS–CVX).** Static-PCI earns +37.7%, +20.5% and +19.6% with low drawdown, Dynamic-PCI is mainly near zero or negative, and Neural KalmanBOT matches or exceeds the static finding (+40.2%, +40.1%, +32.4%), all things considered; and thus, its Sharpe on XOM–CVX (0.98) is the highest in the study.
- **Static hedge damaging (NKE–AES, HD–LOW, KO–PEP).** Static-PCI loses 64.4%, 21.4% and 9.1%; Dynamic-PCI limits but doesn't remove the losses; Neural KalmanBOT is positive on all three (+45.7%, +6.8%, +3.9%), on the whole. NKE–AES accounts for most of the total Neural advantage.
- **Commodity and energy pairs (GLD–SLV, USO–XLE, XLE–XOP).** Mainly negative. Gating cut the maximum drawdown on GLD–SLV (−6.5% vs. −25.5% Static) and XLE–XOP (−0.6% vs. −20.4%); Neural is normally worst on GLD–SLV (−27.5%, Sharpe −1.26), at least for now.

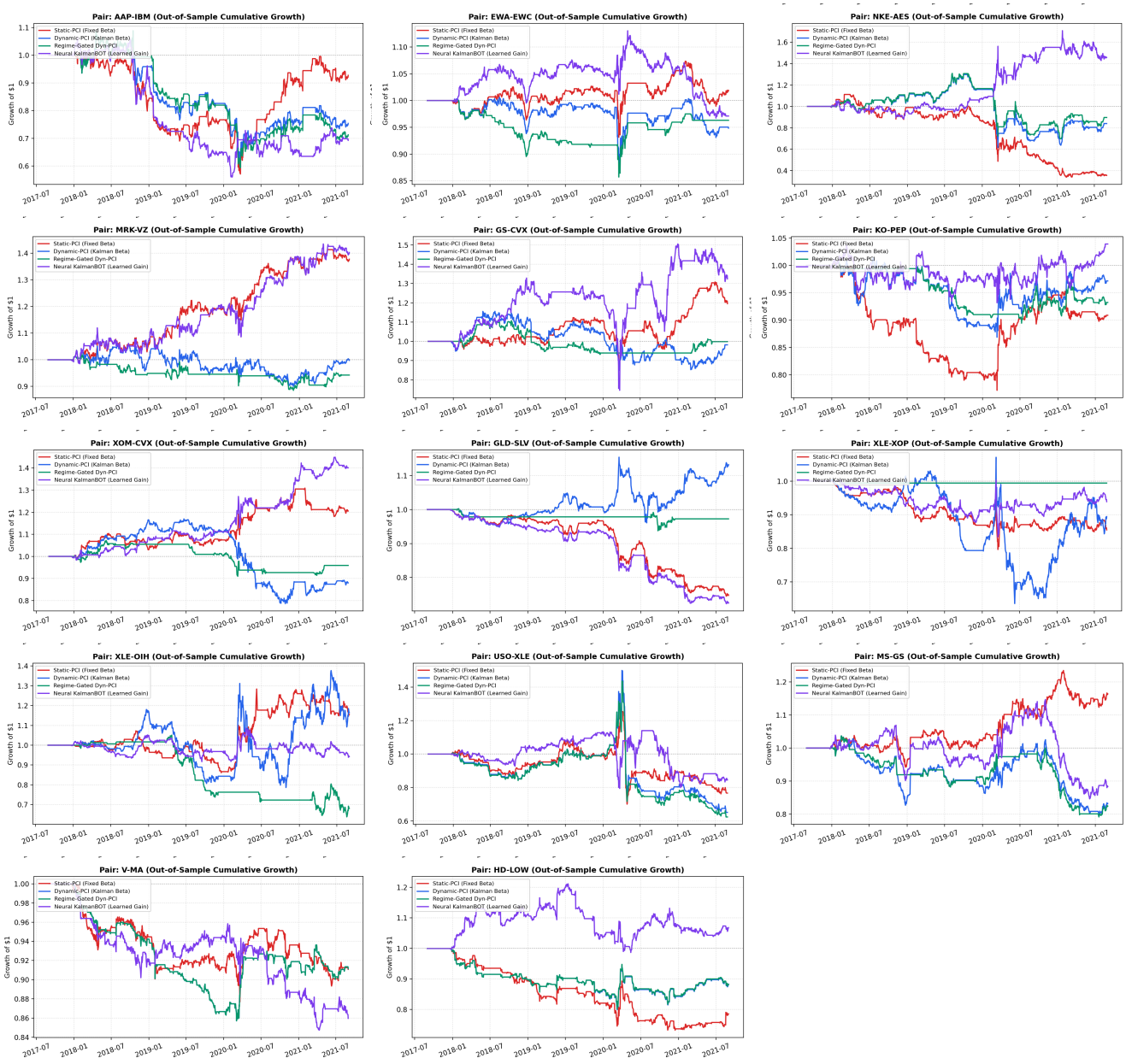


Figure 12: Out-of-sample equity curves (growth of $1) for all 14 pairs and four models, 2017–2021.

- **V–MA, MS–GS, XLE–OIH, EWA–EWC.** No model is, more accurately, consistently (specifically in this setting) profitable; in turn, all four are losing on V–MA (−8.8% to −14.0%), and Gated equals Dynamic there, in many respects.

The Regime-Gated model is the principal negative finding, including no positive return on any pair and the lowest mean Sharpe (−0.35), in many respects, but still, it lessened maximum drawdown relative to Dynamic-PCI on 9 of 14 pairs, at least for now.

9 Formation-Window and Execution-Rule Ablations

Two design choices were pored over with the Dynamic-PCI model: *how much history* to use for estimation, and *which entry/exit rule* to use, broadly speaking, and both studies are exploratory - they're using four pairs each, one out-of-sample period (2017–2021) and 10 bps costs, and several comparisons were made after seeing the findings, in actual application.

Key takeaways

- No formation window is typically profitable on average; 2 years is the least bad (−5.0%) but this rests on one pair (KO–PEP), at least in principle.
- Very short (1 year) and very long (8 years) windows unstably estimate ρ and the half-life.
- On all four pairs, the Clegg–Krauss expanding-window z-score beat the rolling 80-day rule.
- The unconditional z-score has the best Sharpe ratios but trades only 1–5 times, so those Sharpe ratios aren't trustworthy, at least in practice.

9.1 Formation-Window Length

The formation window was set to 1, 2, 3, 5 and 8 years (253 to 2,014 days) for KO–PEP, EWA–EWC, V–MA and HD–LOW, at least in practice. As, well, the model was then traded over the same out-of-sample period.

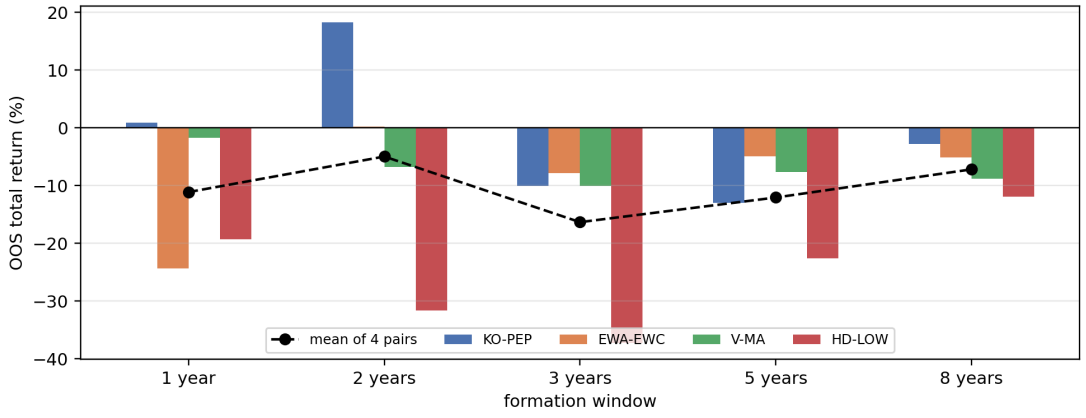


Figure 13: Out-of-sample total return of Dynamic-PCI by formation-window length. The dashed line is the mean over the four pairs.

Reading.

- Fitted factors are, after all, unstable across the windows, like the KO–PEP half-life jumps between 4 and 693 days and several approximations are generally degenerate (HL = ∞),

	Pair	1 yr	2 yr	3 yr	5 yr	8 yr
OOS total return (%)	KO–PEP	+0.92	+18.34	−10.11	−13.00	−2.83
	EWA–EWC	−24.35	+0.21	−7.81	−4.93	−5.16
	V–MA	−1.73	−6.73	−10.11	−7.67	−8.84
	HD–LOW	−19.30	−31.66	−37.28	−22.63	−11.90
	Mean	**−11.1**	**−5.0**	**−16.3**	**−12.1**	**−7.2**
Estimated half-life (d)	KO–PEP	20.6	692.8	10.5	4.2	692.8
	EWA–EWC	3.0	52.5	18.6	74.8	136.4
	V–MA	692.8	120.0	80.7	4.5	1.0
	HD–LOW	0.9	∞	∞	3.1	2.0

Table 9: Out-of-sample return and formation-window half-life of Dynamic-PCI by window length.

the boundary behaviour of Proposition 1.5, at least in practice.

- A 1-year window (about 250 remarks for five coefficients) gave −24.4% on EWA–EWC; an 8-year window averages over several regimes and $\hat{\rho}$ drifts to 0.999 (as reported in the project analysis), all things considered.
- The 2-year mean is, after all perhaps, the least negative; that said, is usually driven by KO–PEP; choosing a window from out-of-sample findings would be data snooping, so cross-validation or walk-forward re-estimation is normally needed, in actual application.

9.2 Entry and Exit Rules

All trade the filtered $\hat{M}_t$. Five ways of forming the trading z-score were compared (Table 10). The unconditional z-score is

Rule	z-score normalisation	Entry	Exit
Standard (production)	rolling 80-day mean/std	$\lvert z\rvert \geq 1.0$	$z = 0$ crossing
Conservative	rolling 80-day	$\lvert z\rvert \geq 1.5$	$z = 0$ crossing
C&K expanding [1]	all data up to t	$\lvert z\rvert \geq 1.0$	$z = 0$ crossing
Half-exit	rolling 80-day	$\lvert z\rvert \geq 1.0$ or 1.5	$\lvert z\rvert \leq 0.5$
Unconditional	$\sigma_M/\sqrt{1-\rho^2}$ (AR(1) stationary std)	$\lvert z\rvert \geq 1.0$	$z = 0$ crossing

Table 10: Definitions of the five execution rules.

$$z_t^{\text{uncond}} = \frac{\hat{M}_t}{\sigma_M/\sqrt{1-\rho^2}}. \tag{26}$$

Reading.

- The expanding window beats the standard rolling rule on return in all four pairs; the project analysis is attributing this to rolling windows adjusting to a supported dislocation and exiting early, in the main.
- The unconditional z-score has the highest Sharpe ratios but only 1–5 trades; with $\hat{\rho} \approx 0.999$ its denominator is usually about $22\,\sigma_M$, so those Sharpe ratios rest on very few notes, to a significant degree.
- Half-exit is helping XOM–CVX but hurts MRK–VZ and EWA–EWC.
- Four pairs, one period, rules assessed after seeing findings, such as the ranking is currently exploratory.

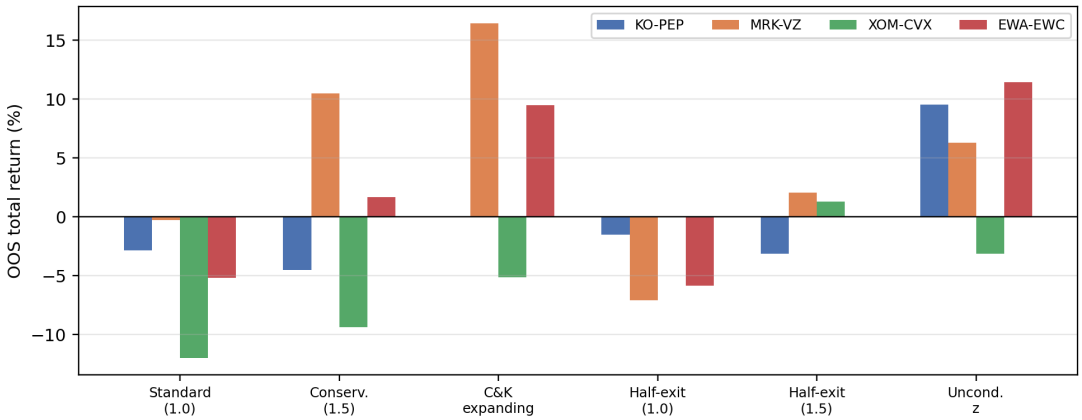


Figure 14: Out-of-sample total return of Dynamic-PCI under each execution rule (missing bars: combination not reported).

	Pair	Standard	Conserv.	C&K exp.	Half-exit 1.0	Half-exit 1.5	Uncond.
Total return (%)	KO–PEP	−2.83	−4.49	+0.06	−1.50	−3.11	+9.51
	MRK–VZ	−0.25	+10.50	+16.44	−7.06	+2.04	+6.30
	XOM–CVX	−11.99	−9.35	−5.14	–	+1.31	−3.12
	EWA–EWC	−5.16	+1.70	+9.49	−5.86	–	+11.44
Sharpe	KO–PEP	−0.04	−0.10	0.04	−0.00	−0.06	0.35
	MRK–VZ	0.05	0.29	0.38	−0.10	0.10	0.29
	XOM–CVX	−0.23	−0.18	−0.05	–	0.08	−0.01
	EWA–EWC	−0.18	0.10	0.41	−0.19	–	0.71
# trades	KO–PEP	22	20	15	21	19	4
	MRK–VZ	23	23	14	19	19	2
	XOM–CVX	41	35	17	–	29	1
	EWA–EWC	41	30	27	35	–	5

Table 11: Execution-rule comparison for Dynamic-PCI, 2017–2021, 10 bps costs ("–": not reported).

10 Failure Modes and Root-Cause Analysis

What the development history included was a few defects whose diagnosis raised the final framework, in the main.

1. **Cascaded (decoupled) filtering.** Estimating β_t by one filter and decomposing residuals with another supposes white residuals, going against the PCI structure in which residuals include an AR(1) and a random walk, and the point estimate $\hat{\beta}_t$ passed on discards its posterior variance, compounding lag, in actual application; the joint filter is avoiding this.
2. **Uncoupled 4-state formulation.** The stable form treats $X_{1,t}$ as an exogenous regressor in $\boldsymbol{H}_t$ (the 3-state model; see Section 2.3), to a significant degree, and an early version modelled $X_{1,t}$ with no observation row and an unspecified process variance, inflating $\boldsymbol{Q}$ and creating feedback loops, at least in practice.
3. **Transition-matrix index transposition.** Assigning ρ to the β entry and 1 to the M entry made β_t mean-revert to zero, in real-world scenarios; meanwhile, M_t behaved as a random walk, exactly inverting the model.
4. **GRU hidden-state disconnection.** A function-signature mismatch passed the innovation variance into the position intended for $\boldsymbol{h}_{t-1}$, so the GRU was re-initialised every step and had no memory during BPTT, in actual application.

5. **Premature exit threshold.** Closing at -0.5σ rather than at the zero crossing lost most of the mean-reversion profit, on the whole.
6. **Gradient explosion and `NaN` losses** in BPTT (Section 5.4), settled by clamping, capacity reduction, gradient clipping and detaching.

The seventh issue discovered during the preparation of this report is a *plotting* defect: the whitening panel in the original validation figure was mis-scaled (Section 7); the quantitative findings were unaffected, and the panel was replaced, at least in practice.

11 Discussion, Limitations and Future Directions

11.1 What the Evidence Shows

- On copied data β_0 and ρ are typically recovered to about 1% and innovations are mainly white and normal at true factors (Section 7), at least empirically.
- A static hedge ratio is often damaging on pairs with drift (Static-PCI: -64.4% on NKE–AES, -21.4% on HD–LOW); the dynamic and learned-gain models are limiting or reverse these losses, at least in principle.
- The highest mean return ($+3.9\%$) and the lowest mean and worst-case drawdown, but a negative median, 6 of 14 profitable pairs (as many as Static-PCI), and gains packed together in a few pairs are had by Neural KalmanBOT, in actual application.
- The calibrated screen separates winners from losers well on this sample (12/14); so, though, its thresholds were tuned after the fact, on the whole.

11.2 Limitations

1. **Short test window.** Under an i.i.d. approximation the standard departure of an annualised Sharpe ratio is about 0.5 (10); no significance tests were run on the model comparisons, in the main. About four years with one regime shift (2020).
2. **Single-run neural results.** As, well, the optimiser, learning rate and seed aren't recorded here, so no multi-seed ensemble or validation split is reported, in many respects.
3. **Post-hoc screen calibration** (Section 3).
4. **Test-statistic caveats.** ADF on estimated spreads is employing standard critical values, and plus, aDF on a filtered state isn't a standard test, at least in principle.
5. **Costs and pair selection.** A flat 10 bps ignores spread fluctuation and borrow costs; the pairs are generally well-known tickers present, throughout the sample, in real-world scenarios.

11.3 Future Research

1. **Walk-forward engine:** 2-year formation and 6-month trading windows with continuous re-estimation, and walk-forward validation of the screen, in actual application.
2. **Multi-seed neural ensembles** with validation-based early stopping, reporting dispersion of findings and formal tests of the learned-gain advantage, at least in practice.
3. **Monte-Carlo identification study:** bias and variance of $\hat{\sigma}_M, \hat{\sigma}_R, \hat{\rho}$ as functions of ρ and sample size, formalising Proposition 1.5, on the whole.
4. **Log-price observation model** via EKF/UKF for geometric compounding.
5. **Multi-asset baskets:** vector state-space partial cointegration for ETF-sector arbitrage.
6. **Hidden Markov regimes:** an online Hamilton filter [26] for posterior regime probabilities (stationary, stormy, different) to scale positions by $1.0\times, 0.5\times, 0.0\times$ instead of a hard gate, all things considered.

7. **Online parameter tracking:** recursive EM replacing batch L-BFGS-B.
8. **Deflated Sharpe ratios** [24] and block-bootstrap intervals for strategy returns.

11.4 Conclusion

A stabilised 3-state Dynamic-PCI framework was built by the project (and recorded why a 4-state formulation was discarded), obtained the magnitudes employed for screening and diagnostics (R^2_{MR}, half-life, identification, innovation likelihood), prescribed a small GRU-based Kalman-gain network with a two-stage curriculum ending in a cost-aware differentiable Sharpe loss, and benchmarked four filters on 14 pairs over four out-of-sample years, in real-world scenarios. The learned-gain model delivered the highest mean return and the lowest drawdowns and turned several large static-hedge losses into the gains, in real-world scenarios. The proof is, though, restricted by a short sample, single-run neural findings and post-hoc screen calibration; we regard the learned-gain approach as a promising direction whose confirmation is needing walk-forward check, multi-seed ensembles and pre-registered rules.

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Annexure-II

DECLARATION

I/We hereby declare that the work presented in the project report entitled Dynamic PCT & KalmanBOT for pairs trading is written by me in my own words and contains my own or borrowed ideas. At places, where ideas and words are borrowed from other sources, proper references and acknowledgements, as applicable, have been provided. To the best of my knowledge this work does not emanate from or resemble work created by person(s) other than those mentioned and acknowledged herein.

Name and Signature ARYAN DEO

Date: 29/10/26

29/10/2026.